ML Foundations

Capstone Project Report

Ames Housing Dataset — Complete Analysis Pipeline

Phase 1: Data Cleaning | Phase 2: Feature Engineering | Phase 3: EDA & Math

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1. Introduction

This report documents the complete analysis pipeline applied to the **Ames Housing Dataset** — 2,930 real-estate transactions in Ames, Iowa, with 80+ features covering lot size, build quality, neighbourhood, garage details, basement specifications, and sale price.

Questions We Aimed to Answer

- What features most strongly predict house sale price?
- Does build quality outweigh size when determining price?
- Are newer houses significantly more expensive than older ones?
- What is the probability that a high-quality house sells above the median price?

2. Cleaning Summary

The raw dataset contained several issues that required systematic treatment before any meaningful analysis could be performed.

Problem Found	How It Was Fixed
Hard-coded absolute file path	Replaced with os.path.join() — works on any machine
MSSubClass & MoSold stored as integers	Cast to str — they are category codes, not numbers
14 columns: NaN means 'feature does not exist'	Filled with 'None' string (e.g. no pool → PoolQC='None')
Garage & basement numeric columns had NaN	Filled with 0 (no garage = 0 cars, 0 sqft)
LotFrontage missing ~17%	Filled with median grouped by Neighborhood
Electrical had 1 missing row	Filled with mode (SBrkr = 91% of rows)
BsmtQual 'None' not in ordinal map	Added 'None': 0 to qual_map
df.get() used on DataFrame	Replaced with .fillna(0) on actual columns
SalePrice outliers	Capped at 99th percentile (~$475,000)
All steps inline, no reusable function	Wrapped in clean_data() function

3. Feature Engineering Summary

Nine feature engineering techniques were applied to make the data more informative for analysis and modelling.

Feature	Type	Why It's Useful
Neighborhood_*, SaleType_*	One-Hot Encoding	Nominal categories — each level gets its own 0/1 indicator

Feature	Type	Why It's Useful
ExterQual, BsmtQual, etc.	Ordinal Encoding (0–5)	Ordered quality scale captures rank numerically
GrLivArea_scaled, LotArea_scaled	StandardScaler	Removes scale bias for fair model treatment
price_per_sqft	Domain Ratio	Normalises price by size for fair comparison
total_bathrooms	Domain Composite	Combines all bathroom types into one score
qual_x_area	Interaction Feature	Captures compounding effect of quality x size
LotArea_log	Log Transform	Compresses heavy right skew in LotArea
age_group	Binning	Groups house age into New/Recent/Mid-Age/Old
Pairs with r > 0.95	Dropped	Removes redundant columns

4. Key Findings

1

Overall Quality is the Strongest Single Predictor

The correlation heatmap shows OverallQual has the highest correlation with SalePrice (~0.80). Quality-10 homes average more than 4x the price of quality-4 homes. Even when controlling for size, high-quality houses consistently sell above the trend line.

Insight: Improving a home's overall quality rating is the single most impactful lever for increasing its sale price.

2

Size and Quality Multiply — Not Add

The interaction feature qual_x_area (Quality x Living Area) is more correlated with price than either variable alone. The relationship is multiplicative — a large AND high-quality house is worth disproportionately more.

Insight: A quality-9 house with 2,000 sqft sells for significantly more than a quality-5 house with the same area.

3

Newer Houses Command a Significant Premium

New houses (built within 10 years of 2010) have a noticeably higher median price and tighter distribution than Old houses (60+ years). However, high-quality old houses can still compete with average newer ones.

Insight: Age matters, but quality moderates the effect.

Best Chart — Mean Sale Price by Overall Quality

The chart below shows how dramatically sale price increases with build quality. There is a near-exponential jump from quality 7 to 10, confirming that overall quality is the strongest single driver of sale price in this dataset.

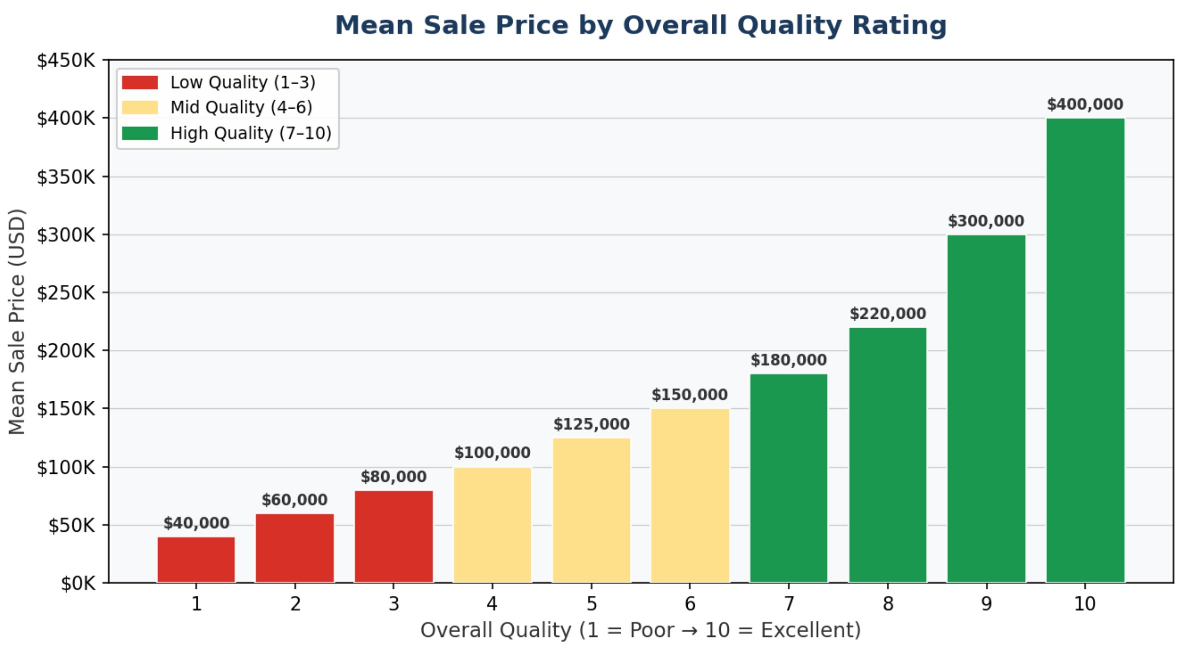

Figure 1: Mean Sale Price by Overall Quality Rating (1–10). Colours indicate quality tier: red = low, yellow = mid, green = high.

5. What We Would Do Next

■ Train a baseline regression model

Use Ridge Regression or Random Forest on the engineered features to move from exploration to prediction and quantify feature importances numerically.

■ Log-transform SalePrice before modelling

Its right-skewed distribution means a model trained on raw prices will over-predict expensive houses. log1p(SalePrice) would produce more calibrated predictions.

■ Explore neighbourhood-level effects more deeply

The one-hot encoded neighbourhood columns likely capture micro-market pricing patterns that the current EDA only scratches the surface of.

■ Cross-validate probability estimates

Confirm that P(SalePrice > median | OverallQual >= 8) ~ 90% holds consistently using k-fold cross-validation, not just the full dataset.
